

NLP Stock Sentiment Analysis: A Comparative Embedding-Based Model Report

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Abstract

This work presents an NLP based framework that converts daily financial news into structured sentiment scores to help investors detect market mood and anticipate price movements. The system transforms news headlines into vector representations using Word2Vec, GloVe, and sentence level embeddings, followed by sentiment classification using a Gradient Boosting model. Weekly sentiment summaries are generated and aligned with historical OHLCV data to illustrate how textual indicators can complement market analysis. By combining unstructured text with market metrics, the framework provides a simple and reproducible way to monitor investor perception and evaluate trends. All code and model artifacts are publicly available for reuse.

Keywords

Sentiment Analysis, Financial News, Word2Vec, GloVe, SentenceTransformer-based Embedding, Gradient Boosting, NLP Pipeline, Stock Market Analysis, Embedding Comparison, Text Classification

1. Introduction

Financial markets respond strongly to how investors interpret news, and shifts in sentiment often precede changes in price. However, analysts and investment teams frequently rely on manual reading or intuition when assessing news flow. This process is slow, subjective, and inconsistent. Automated sentiment analysis offers a consistent and scalable approach for transforming large volumes of text into structured numerical signals that can support market evaluations.

This paper develops a reproducible sentiment analysis pipeline that processes financial news, converts headlines into embedding-based vector representations, and classifies sentiment using a classical machine learning model. No transformer fine-tuning or deep neural architectures are used in this study. The objective is to provide a clear demonstration of how traditional NLP techniques can support investment research by offering early signals of market mood.

2. Business Context

Investment decision making depends not only on technical indicators and price movement but also on how the market interprets ongoing events. Analysts want early detection of changes in sentiment, especially when news volume is high. An automated sentiment scoring system can support weekly market reviews, research summaries, and risk assessments by quantifying whether news within a given period is broadly positive, neutral, or negative. This provides a stable and repeatable method for understanding market perception.

3. Problem Statement

Analysts need a reliable way to quantify sentiment from financial news so that they can track changes in market mood and relate these changes to price movement. Manual review is inconsistent and does not scale, and without structured numerical sentiment values it is difficult to analyze trends over time. The goal of this work is to create an automated pipeline that classifies daily news sentiment, aggregates it over time, and pairs these results with market data to demonstrate how such sentiment indicators can support evaluation of market conditions.

The goal is to deliver a simple and efficient classical machine learning pipeline that operates effectively on compact datasets without relying on large transformer-based models.

4. Data Source

Origin. This study uses a curated dataset that pairs daily financial news headlines with corresponding OHLCV market indicators. The dataset was compiled specifically for this sentiment analysis experiment and is publicly available on Zenodo for transparency and reproducibility. The complete dataset, along with documentation, is accessible at [\[1\]](#).

Structure. The dataset is provided in a single CSV file that contains the columns listed in Table 1. Each record includes the trading date, the headline text, standard OHLCV market variables, and a sentiment label. The sentiment annotation uses the values 1 for positive, 0 for neutral, and minus 1 for negative sentiment.

Column	Description
Date	Trading date associated with the news headline
News	Text of the headline or news snippet
Open	Opening price of the asset
High	Highest price of the day
Low	Lowest price of the day
Close	Adjusted closing price
Volume	Daily traded volume
Label	Sentiment label where 1 is positive, 0 is neutral, and minus 1 is negative

Table 1. Dataset Column Descriptions

Size and Properties. Table 2 summarizes the primary characteristics of the dataset. A total of 349 headlines are included, with one entry per trading day. The text samples are short to medium in length and span a continuous time window. The dataset contains minimal missing data and provides sentiment labels required for supervised learning.

Size	Value
Records	349
Sampling	One record per trading day
Text length	Short or medium
Coverage	Continuous date range
Missing Data	Minimal
Labels	Provided for supervised learning

Table 2. Dataset Size and Properties

Why this Dataset is Suitable. This dataset is well suited for demonstrating a complete sentiment analysis workflow because each row pairs unstructured text with structured market indicators, enabling direct comparison between sentiment and stock behavior. The continuous daily sequence allows chronological evaluation and weekly aggregation. Although compact, the dataset provides a clean and consistent foundation for illustrating preprocessing, embedding generation, model training, evaluation, and sentiment aggregation in a reproducible manner.

5. Exploratory Data Analysis (EDA)

Sentiment Distribution. The dataset contains headline-level financial news samples labeled as Positive, Neutral, or Negative. Neutral entries dominate the distribution, while Positive and Negative samples appear in comparable proportions. This slight imbalance reflects the factual nature of financial reporting and indicates that models must handle a relatively neutral-heavy dataset.

Headline Length Characteristics. Each headline is short to medium in length; a statistical summary of word counts shows a compact range without extreme outliers, which supports efficient tokenization and embedding without aggressive truncation. The dataset is complete, with no missing values or duplicate records, and manual inspection of samples indicates that sentiment labels appear coherent with the headline context.

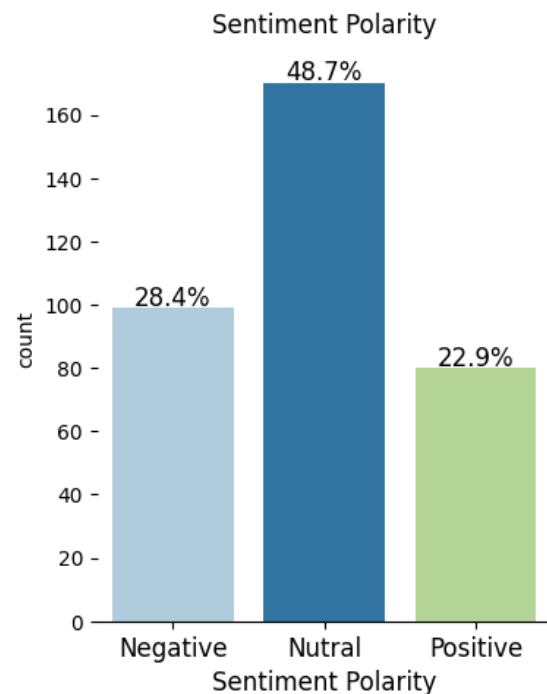


Figure 1. Sentiment Distribution of Financial News Dataset

Price and Volume Context (Not Used for Feature Engineering). Although the raw dataset includes OHLC prices and trading volume, these numeric variables are not directly used in the sentiment classifier. Their distributions were reviewed only to confirm data completeness and validate that the news spans normal market conditions. These variables were excluded to maintain a text-focused sentiment modeling approach.

Overall Assessment. The dataset is clean, complete, and well suited for embedding-based classification using Word2Vec, GloVe, and sentence-level embeddings.

6. Embedding Methods

Word2Vec Embedding (300-dimensional). Word2Vec was used to generate continuous vector representations of each headline by training a Skip-Gram model on the available corpus. The embedding dimension was set to 300 to capture semantic relationships between words. Document-level embeddings were obtained by averaging the word vectors in each headline, which provides a stable representation for short text samples. Because the dataset is compact, the semantic richness of the Word2Vec model is limited, and this embedding variant is included primarily for comparative evaluation.

GloVe Embedding (100-dimensional). A second embedding variant used pretrained GloVe vectors with a 100-dimensional representation. Each headline was transformed by averaging the GloVe vectors of its constituent words. GloVe provides high-quality global semantic structure learned from large external corpora, which makes it suitable for text classification tasks where the dataset itself is not large enough to support training domain-specific embeddings.

SentenceTransformer-based Embedding (384-dimensional). To obtain richer sentence-level representations, the all-MiniLM-L6-v2 encoder from the SentenceTransformer library was applied to each headline. This model produces a 384-dimensional dense embedding for the entire sentence without requiring any fine-tuning. The encoder provides compact sentence representations that capture semantic similarity across headlines while keeping computational cost low.

Embedding Summary. These three embedding strategies represent different levels of semantic abstraction. Word2Vec and GloVe provide word-level embeddings aggregated through averaging, while the SentenceTransformer-based encoder produces unified sentence-level vectors. Each embedding set is evaluated independently using identical classification pipelines to compare the impact of representation quality on sentiment prediction.

7. Model Architecture and Training Setup

Gradient Boosting Classifier. All experiments used a Gradient Boosting Classifier as the predictive model. Gradient boosting constructs an ensemble of weak learners, typically decision trees, where each subsequent learner focuses on correcting residual errors from earlier ones. This method is effective for medium-sized datasets and provides strong performance without requiring deep neural architectures. Its ability to model nonlinear patterns while maintaining interpretability makes it a suitable baseline for evaluating embedding-based text representations.

Base Models. For each of the three embedding types Word2Vec, GloVe, and SentenceTransformer-based sentence embeddings a baseline model was trained using default hyperparameters. These models establish a reference point for evaluating how well each embedding captures sentiment-related information. Because the embeddings differ in dimensionality and semantic richness, the baseline results help illustrate the direct impact of representation quality on classification outcomes.

Tuned Models via Grid Search. To improve performance over the baseline, hyperparameters were optimized using GridSearchCV with five-fold cross-validation. The grid search explored combinations of the number of estimators, learning rate, and maximum tree depth. These parameters control the complexity and learning dynamics of the ensemble. The configuration that achieved the highest validation F1 score for each embedding type was selected as the tuned model. This ensures that the comparison across embedding variants reflects their best attainable performance within the same model family.

Training Procedure. The dataset was split into an 80 percent training set and a 20 percent validation set using a fixed random seed to ensure reproducibility. All embeddings were computed prior to model training so that each classifier operated on a consistent feature representation. Each model was trained and evaluated on the same data split to maintain fairness in comparison. Performance was assessed using accuracy, precision, recall, F1 score, and error rate, providing a balanced view of model behavior across sentiment classes.

Model Comparison. Both the baseline and tuned models were evaluated for each embedding type. This comparison highlights the effect of hyperparameter optimization and allows direct assessment of how representation quality influences performance. Because the dataset is small, all models achieve perfect training accuracy, making validation metrics the primary indicator of true generalization capability.

8. Results and Model Comparison

Three embedding-based models were evaluated using the Gradient Boosting Classifier. Each model was trained on its respective vector representation and assessed on a held-out validation set. All base models achieved perfect accuracy on the training data, which indicates overfitting and highlights the importance of focusing on validation metrics when evaluating generalization.

The Word2Vec-based model showed weak validation performance, with low accuracy and a high error rate. This outcome is expected because the Word2Vec embeddings were trained on a very small corpus, which limits their semantic richness and reduces their ability to capture sentiment in financial text. The GloVe-based model performed more consistently and achieved higher validation accuracy than Word2Vec, benefiting from pretrained vectors derived from a much larger external corpus.

The SentenceTransformer-based sentence embeddings delivered stronger baseline performance than Word2Vec and GloVe, reflecting their ability to capture broader sentence-level semantics. However, validation performance still exhibited variability across classes, consistent with the compact dataset size.

Hyperparameter tuning improved results across all embedding types. Among all candidates, the tuned GloVe model achieved the strongest validation performance, with the highest accuracy, recall, precision, and F1 score, as well as the lowest error rate. This suggests that global co-occurrence statistics combined with simple mean-pooled representations are well suited to short headline-style text.

The complete set of performance metrics is reported in Table 3, which provides accuracy, recall, precision, F1 score, and error rate for each model on both the training and validation sets.

Model	Set	Accuracy	Recall	Precision	F1 Score	Error
Word2Vec – Base Model	Training	1	1	1	1	0
	Validation	0.381	0.381	0.381	0.371	0.667
Word2Vec – Tuned Model	Training	1	1	1	1	0
	Validation	0.381	0.381	0.269	0.315	0.619
GloVe – Base Model	Training	1	1	1	1	0
	Validation	0.476	0.476	0.406	0.438	0.571
GloVe – Tuned Model	Training	1	1	1	1	0
	Validation	0.714	0.714	0.758	0.694	0.286

Table 3. Performance Comparison of Base and Tuned Models Across All Embedding Methods

On the held-out test set, the tuned GloVe model achieved an accuracy of 0.429, recall of 0.429, precision of 0.392, an F1 score of 0.388, and an error rate of 0.619. This drop relative to validation performance further confirms the expected limitations caused by data scarcity and class imbalance.

Overall, the results demonstrate that embedding quality has a significant impact on sentiment prediction performance, and among the methods evaluated, the tuned GloVe model is the most effective for headline-level financial sentiment classification in this dataset.

9. Weekly Sentiment Aggregation and News Summarization

After evaluating model performance, the dataset was aggregated on a weekly basis to observe broader sentiment patterns and to generate summarized narratives of market activity. The original dataset contains one headline per trading day, which was grouped into weekly intervals using calendar-based aggregation. All headlines within each week were combined to form a single text block representing the news flow for that period.

The concatenated weekly text was then passed to a large language model (Mistral-7B-Instruct) to generate a short narrative summary describing the main themes, events, and tone of the market for that week. This summarization step serves as a downstream application that demonstrates how sentiment classification and text aggregation can support market commentary, research workflows, or automated reporting systems. A total of eighteen weekly summaries were created, covering the complete time span of the dataset.

The combined approach of weekly sentiment scoring and LLM-based summarization illustrates how embedding-driven classification can be extended beyond discrete labels to produce higher-level insights. While the sentiment classifier quantifies whether news in a given week is predominantly positive, neutral, or negative, the summarization step provides contextual interpretation that complements these numerical indicators. Together, these outputs show how automated NLP techniques can be used to support market analysis and generate concise, consistent summaries of financial news cycles.

10. Conclusion

This study compared three embedding methods for sentiment classification of financial news headlines using a Gradient Boosting classifier. Although all models overfit the training set due to the small dataset size, their validation results reveal important differences in generalization. The tuned GloVe model achieved the strongest overall performance, demonstrating that global co-occurrence embeddings are well suited for short headline-level text. The weekly aggregation and summarization steps show how numerical sentiment outputs and textual summaries can support broader market-oriented analysis. Overall, the pipeline provides a practical and reproducible framework for extracting sentiment signals from financial news.

11. Limitations

The primary limitation of this study is the small dataset, which restricts semantic coverage and contributes to overfitting. Headlines also provide limited context compared to full articles, making sentiment labeling more ambiguous. Performance declines on the test set confirm limited generalization under data-scarce conditions. The summarization component relies on a single LLM configuration, which may introduce stylistic bias. The study does not include larger corpora or domain-specific fine-tuning, which could improve representational quality.

12. Future Work

Future work may expand the dataset to improve generalization and enable richer semantic learning. Pretrained domain-specific models or fine-tuned sentence encoders could be evaluated to determine whether they outperform static embeddings. Incorporating market movement data could support predictive modeling rather than classification alone. Cross-market validation and experimentation across different asset classes would help assess robustness. Enhancing the summarization step with sentiment-weighted or event-driven aggregation may yield more informative market insights.

Appendix A. Supplementary Materials

The supplementary dataset, code notebook, and project documentation are available in the Zenodo archive (DOI: 10.5281/zenodo.17510735).

The model resources and embedding outputs are hosted on Hugging Face at https://huggingface.co/joyjitroy/Stock_Market_News_Sentiment_Analysis.

References

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